

WS03D: Power Dissipated in a Resistor

Time: 50 min

Name: ()

Class: **26S**
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Date:

The circuit in Fig. 1 is set up to investigate the power dissipated in a resistor.

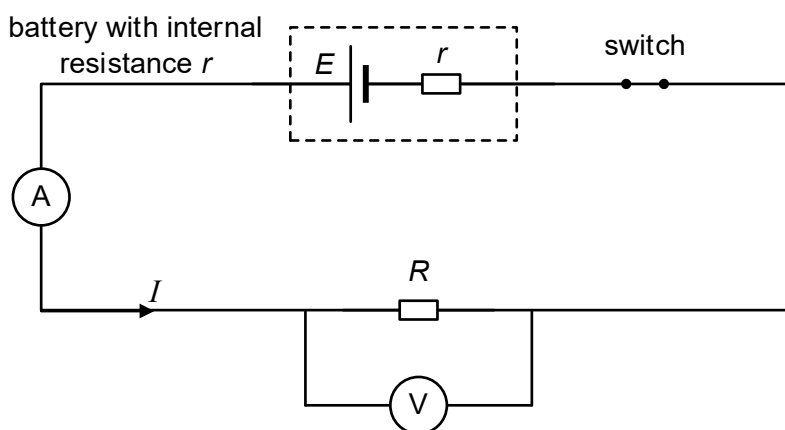


Fig. 1

Theory suggests that the power P dissipated in a resistor and potential difference V across it are related by the expression

$$P = kV^n$$

where k and n are constants.

Linearise the equation

$$P = kV^n$$

to the form $y = mx + c$.

$$\lg P = n \lg V + \lg k$$

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The potential difference V across a resistor is varied and the corresponding current I flowing through the resistor is recorded.

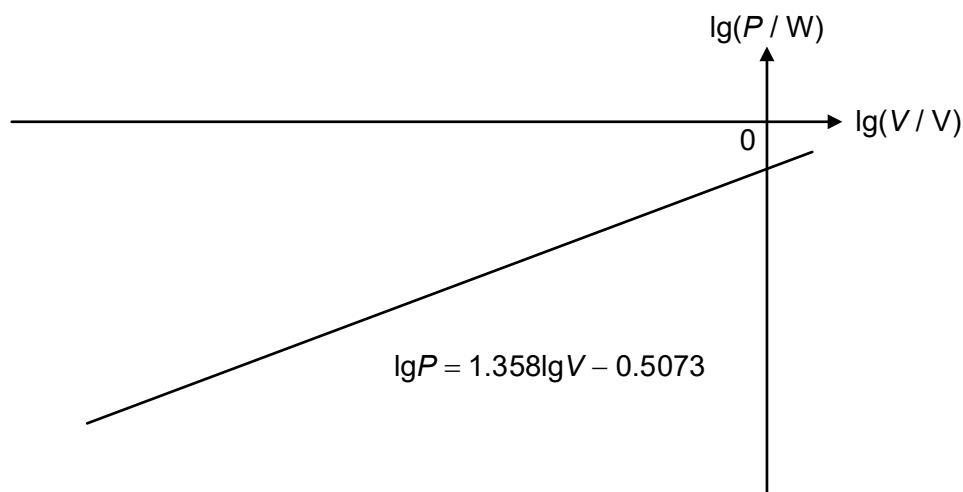
V / V	I / A	P / W	$\lg(V / \text{V})$	$\lg(P / \text{W})$
0.058	0.109	0.0063	-1.24	-2.20
0.092	0.135	0.012	-1.04	-1.91
0.130	0.153	0.0199	-0.886	-1.701
0.429	0.285	0.122	-0.368	-0.913
0.711	0.272	0.193	-0.148	-0.714
1.027	0.308	0.316	0.0116	-0.500
1.321	0.350	0.462	0.1209	-0.335

Additional Guidelines for Tabulation (refer to the guidelines on tabulation on pg. 2 of WS03B).

When logarithms are taken, only numbers to the right of the decimal point (including any zeros) of the logged value are considered significant, i.e., number of s.f. of x = number of d.p. of $\lg x$.

For example, 1.23 is 3 s.f. so the $\lg(1.23)$ should be presented as 0.090.

Sketch the graph in the space below. Write down the equation of the trendline.



Determine the value of k and n , with the appropriate units.

$$k = 10^{-0.5073} = 0.310 \text{ W V}^{-1.358}$$

$$k = 0.310 \text{ W V}^{-1.36}$$

$$n = 1.36$$

Using Excel to Process the Data

Transfer the two columns of raw data (including the headings) to a new Excel sheet.

In C1, type the heading for power dissipated. In C2 to C8, enter the formula to calculate the power dissipated by the resistor (Fig. 2).

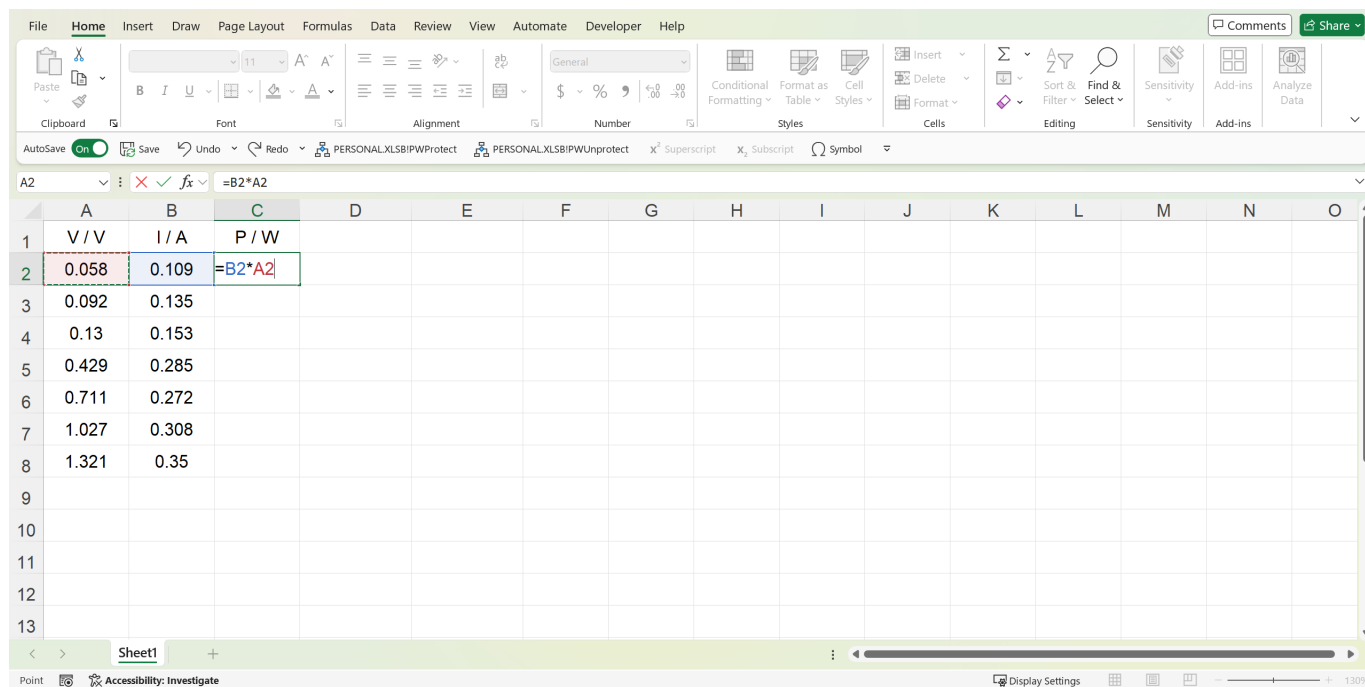


Fig. 2

Based on your linearised equation, compute additional column(s) of data with appropriate heading(s) so that a straight-line graph can be plotted.

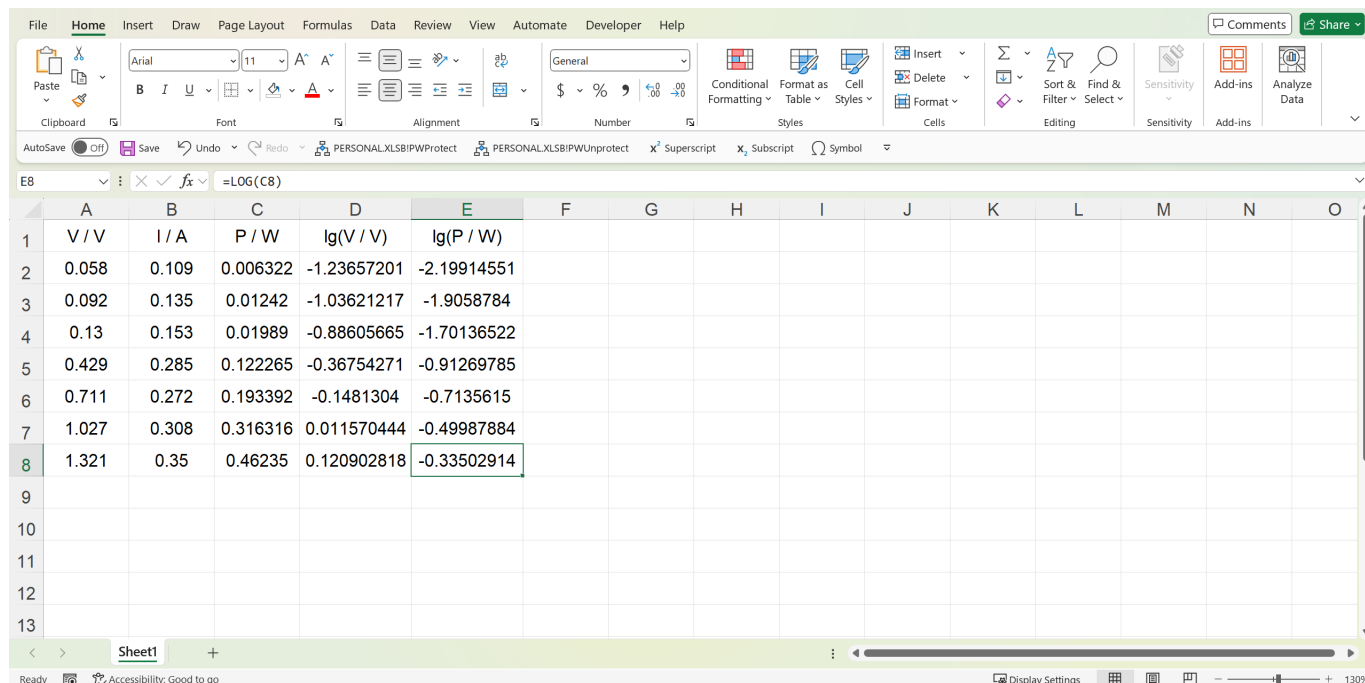


Fig. 3

Transfer the processed data from Excel into the table on pg. 2, with the appropriate number of s.f.

Plot a Graph and Add a Trendline

Select the appropriate columns and insert a *Scatter Chart*.

Ensure that the axes represent the correct y and x quantities.

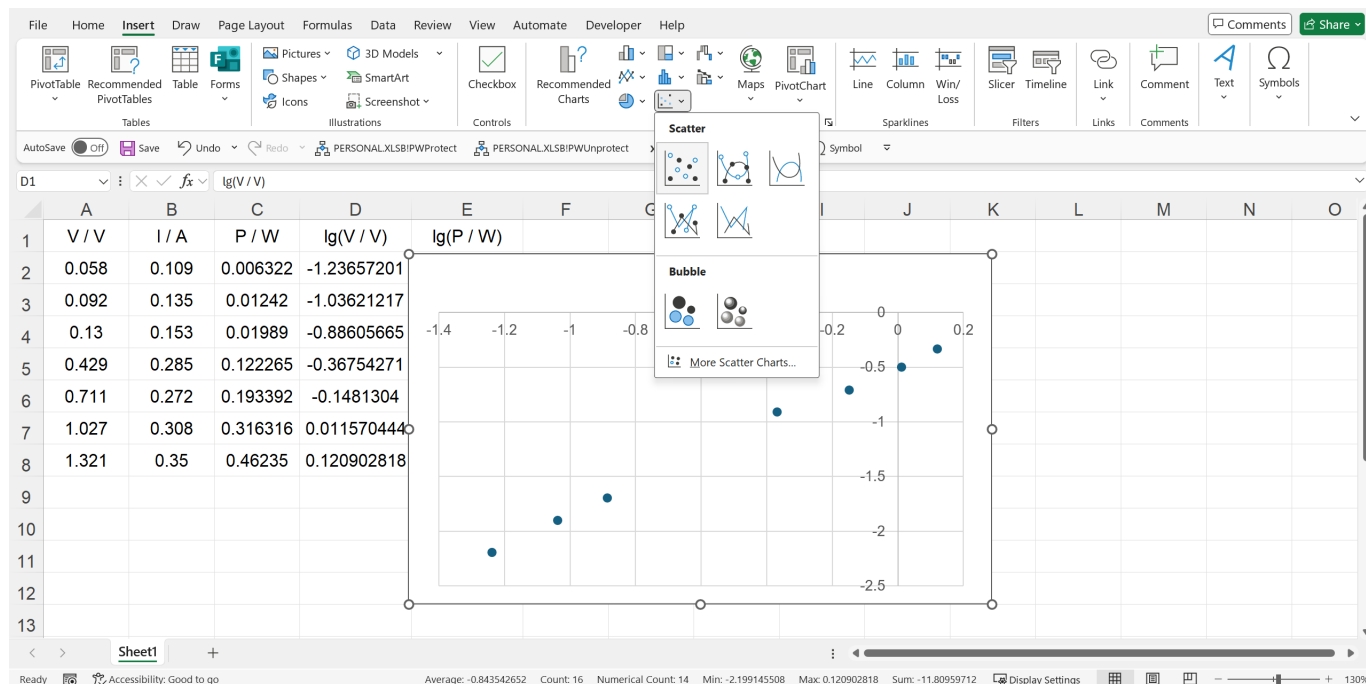


Fig. 4

If the quantities are plotted in reverse, follow these instructions:

Right-click on the chart area and click on *Select Data* [Fig. 5(a)].

Select *Edit* to open the *Edit Series Dialog* [Fig. 5(b)].

Swap/change the *Series X Values* with the *Series Y Values* [Fig. 5(c)].

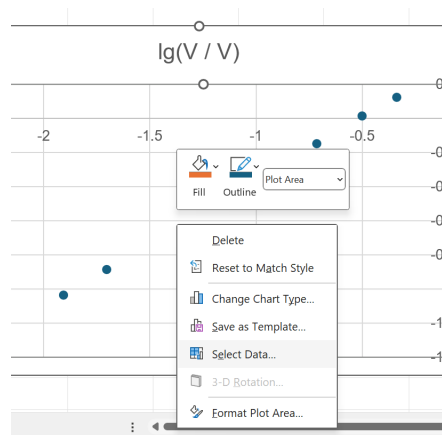


Fig. 5(a)

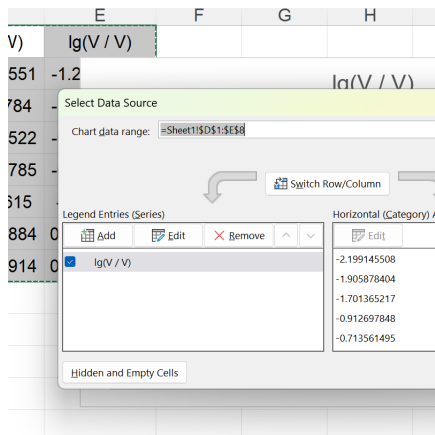


Fig. 5(b)

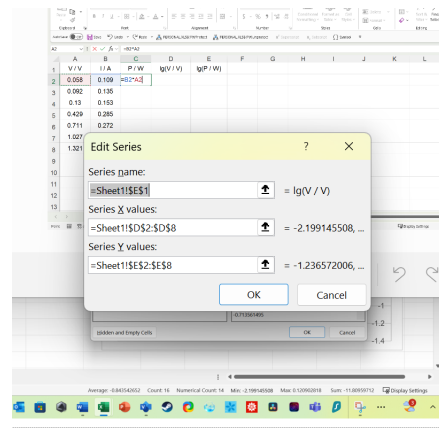
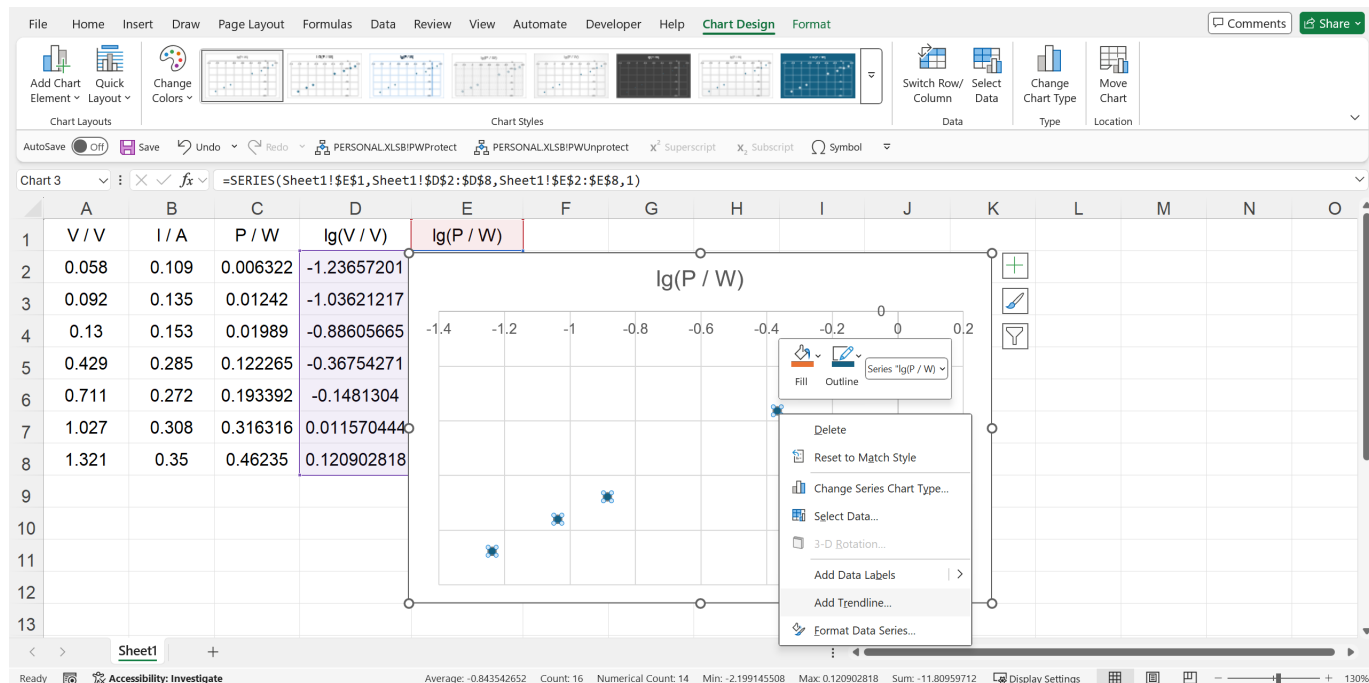


Fig. 5(c)

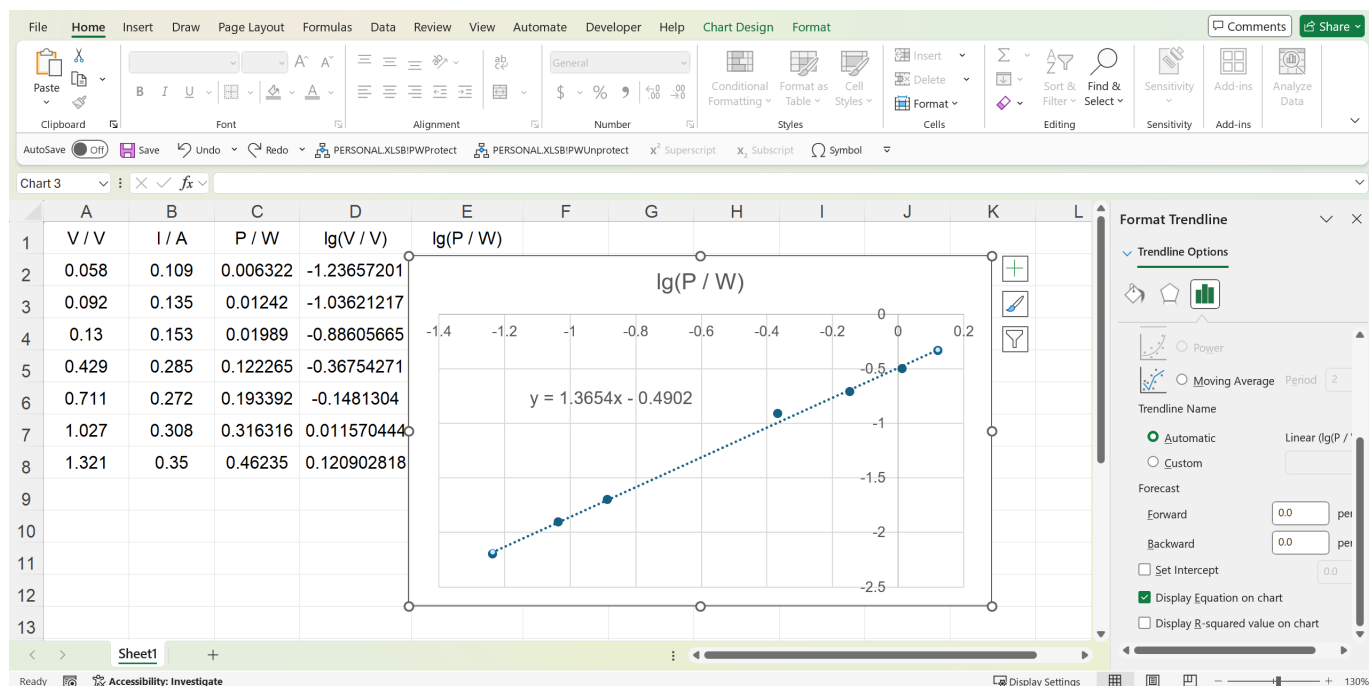
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Right-click on the data and select *Add Trendline*.



In the *Format Trendline* (right panel), check *Display Equation on Chart*.

Move the equation to an empty area on the chart. Increase the font size if necessary.



Dealing with Any Anomalous Point

One point appears to be anomalous as it is relatively far from the trendline compared to other points.

Select the entire Row 5 and hit the *Delete* key. The chart is updated with a new trendline.

Sketch the graph with labelled axes and write down the equation of the trendline on pg. 2.

END OF WORKSHEET